

6 - 25 Ton Unitary Split Systems Indoor

Unit Overview

Model	TWE06044AAA**B100000000000000 000000000000
Unit Tonnage	5 Tons
Max A.H. Operating weight	431.0 lb
Min A.H. Operating weight	232.0 lb
Unit Voltage	460/60/3
Refrigeration Circuit	Single Circuit



Fan Information

Airflow	2000 cfm
External Static Pressure	0.500 in H2O
External Plus Component Static Pressure	0.760 in H2O
Supply fan motor RPM	1094 rpm
Supply fan motor BHP	0.95 bhp

Cooling Information

Airflow	2000 cfm
Cooling EDB	80.00 F
Cooling EWB	67.00 F
Ambient Temperature	95.00 F

Heating Information

Heat Pump Information		Electric Heat Information	
Heating EDB	70.00 F	Heating EDB	70.00 F
Heating Ambient DB	17.00 F		
Heating Ambient WB	15.00 F		

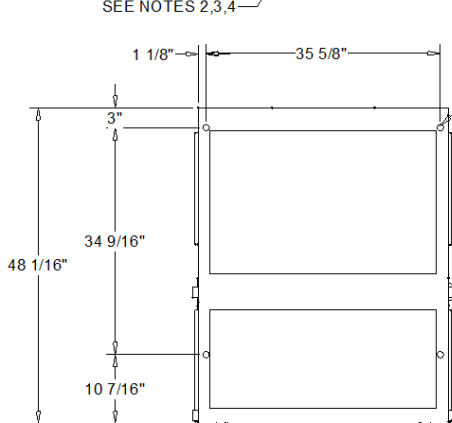
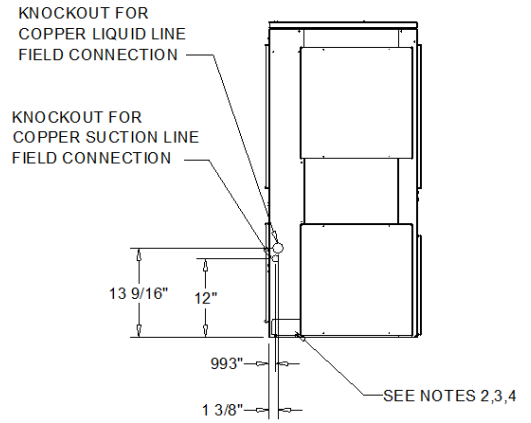
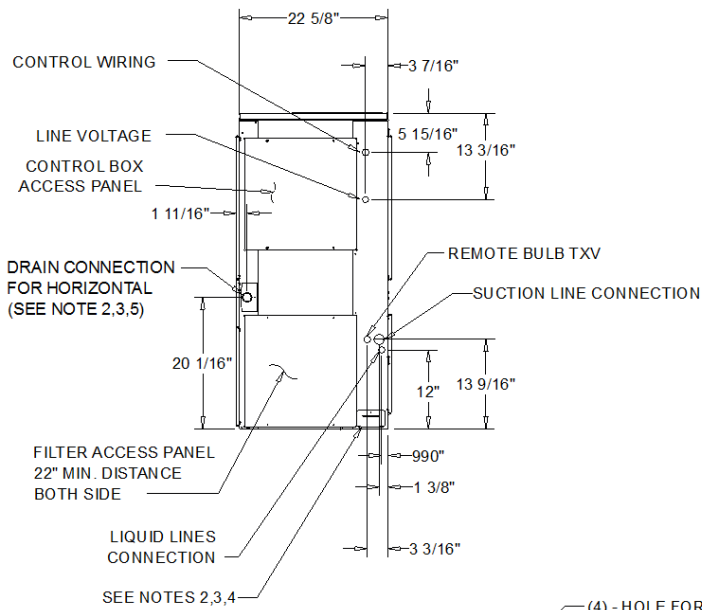
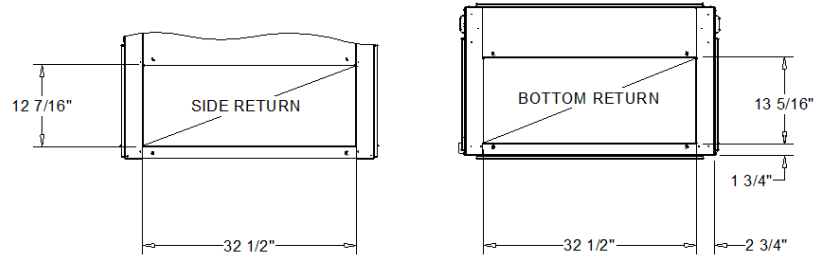
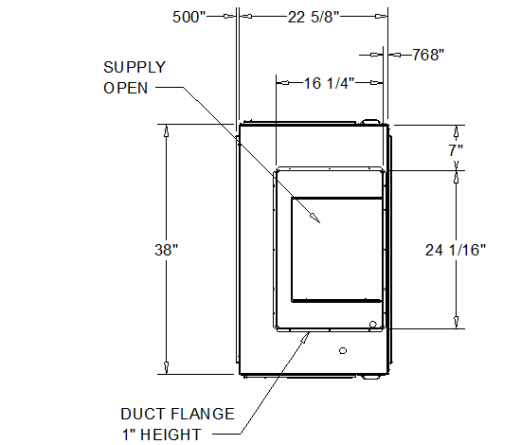
Electrical Information

External Static Pressure	0.500 in H2O	External Plus Component Static Pressure	0.760 in H2O
MCA - A.H.	4.00 A	MCA - A.H. for 230V w/elect. heat	0.00 A
MOP - A.H.	15.00 A	MOP - A.H. for 230V w/elect. heat	0.00 A
Evaporator Motor FLA	1.40 A		

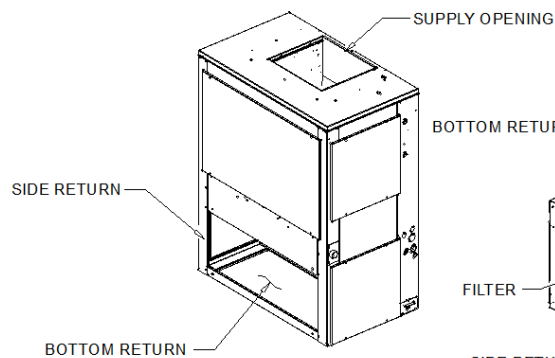
Refrigerant Information

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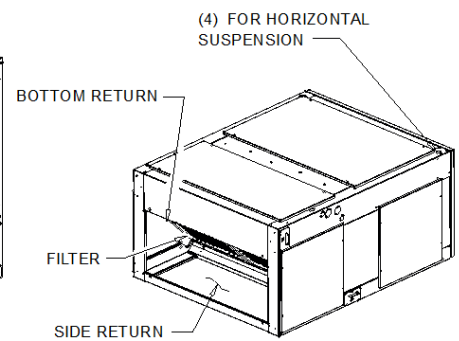
1. PANEL DEPTH 1/2" (TYP. ALL PANELS).
2. REMOVABLE DRAIN PAN AND ATTACHED DRAIN CONNECTION MAY BE INSTALLED ON END OF UNIT IN EITHER THE VERTICAL OR HORIZONTAL CONFIGURATION, PLASTIC DRAIN PAN ACCESS PLATE ON THE END OF UNIT OPPOSITE DRAIN CONNECTION MUST BE REMOVED TO SLIDE DRAIN PAN OUT OF UNIT FOR CLEANING. ACCESS PLATE MUST BE RE-INSTALLED AFTER SLIDING DRAIN PAN BACK INTO UNIT.
3. IF PERIODIC DRAIN PAN CLEANING IS REQUIRED, ALLOW ROOM FOR PARTIAL REMOVAL OF DRAIN PAN CONNECTION AT END OF UNIT.
4. 1" FEMALE SCHED. 40 PVC PIPE DRAIN CONNECTION VERTICAL CONFIGURATION.
5. 1" FEMALE SCHED. 40 PVC PIPE DRAIN CONNECTION HORIZONTAL CONFIGURATION.



(4) - HOLE FOR HORIZONTAL SUSPENSION



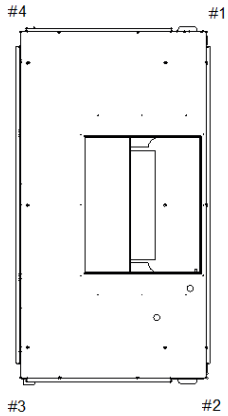
SUPPLY VERTICAL -
 RETURN HORIZONTAL /
 RETURN VERTICAL



SUPPLY HORIZONTAL -
 RETURN VERTICAL /
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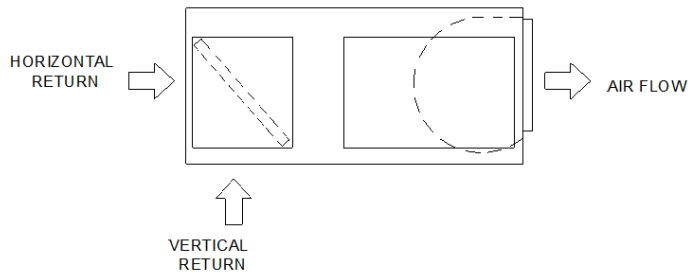
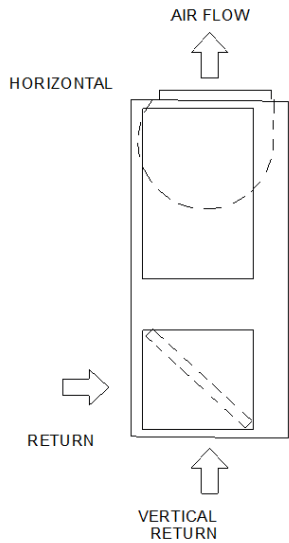
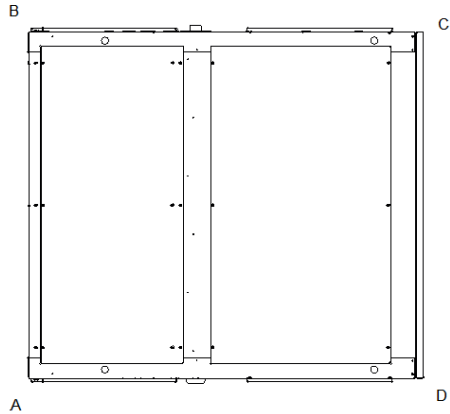
5 TON AIR HANDLER (SINGLE CIRCUIT)

DIMENSIONAL DRAWING



WEIGHTS AND CORNER WEIGHTS

Shipping:	285.0 lb
Net	232.0 lb
VERTICAL	
Corner 1:	55.0 lb
Corner 2:	71.0 lb
Corner 3:	51.0 lb
Corner 4:	55.0 lb
HORIZONTAL	
Corner A:	54.0 lb
Corner B:	67.0 lb
Corner C:	50.0 lb
Corner D:	61.0 lb



WEIGHTS AND LOAD POINT LOCATION FOR CONDENSOR

WEIGHT AND RIGGING



General - (TWE)

- Completely factory assembled
- Convertible for horizontal or vertical configuration
- Convertible for cooling only or heat pump application
- Convertible for left or right external connections (refrigerant and/or electrical)
- Convertible for front or bottom air return
- Nitrogen holding charge
- Certified to UL 1995 for indoor blower coil units

Casing - (TWE)

- Zinc coated, heavy gauge, galvanized steel
- Weather resistant baked enamel finish
- Access panels with captive screws
- Completely insulated with foil faced, cleanable, fire retardant, permanent, odorless glass fiber material
- Captured or sealed insulation edges
- Electrical connection bushings or plugs
- Refrigerant connection bushings or plugs
- Withstand elevated internal static pressure

Refrigeration System - (TWE)

- Single or dual circuit
- Distributor(s)
- Thermal expansion valves (TXVs)

Evaporator Coil - (TWE)

- 3/8" internally enhanced copper tube mechanically bonded to lanced aluminum plate fins
- Factory pressure and leak tested to 449 psig.
- Draw-through airflow
- Dual circuits are interlaced/intertwined
- Double sloped, removable, cleanable, composite drain pan
- Four drain pan positions

Indoor Fan - (TWE)

- Double inlet, double width, forward curved, centrifugal type fan
- Dual fans on 12.5-25 ton air handlers-Adjustable belt drive
- Permanently lubricated bearings

Indoor Motor - (TWE)

- Adjustable motor sheaves (constant volume units)
- Fixed motor sheaves (SZVAV and 2-Speed VFD)
- Thermal overload protection
- Permanently lubricated bearings
- Meet energy policy of 1992 (EPACT)
- Optional oversized motors for high static applications

Controls - (TWE)

- Completely internally wired
- Colored and keyed connectors, colored wires
- Magnetic indoor fan contactor
- Detachable low voltage connectors
- Single point power entry
- Evaporator defrost control

Constant Volume Airflow - (TWE)

- Factory installed high static motor available

Filters - (TWE)

- 2 inch, MERV 13 high efficiency filters



Electric Heaters - (TWE)

- Heavy duty nickel chromium elements
- Agency approved
- Installs directly on fan discharge
- One or two stage control (dependent upon capacity)
- Single point power entry
- Terminal strip connections

230V Heaters

- Internally delta connected
- Automatic reset of high limit controls through pilot duty with secondary backup fuse links

Over Size Motors - (TWE)

- High static applications
- Motor, sheaves, belt included

Condenser and Air Handler Pairings

Table 3. Model number descriptions

TWE Air Handler with Symbio
Digit 15 — Controls 1 = Constant Volume C = 2 Stage Airflow (Electromechanical Condenser Only) D = 2 Stage Airflow/Single Zone VAV (Symbio Condenser Only)
TWE Air Handler (pre-Symbio)
Digit 15 — Controls 0 = Constant Volume A = 2 Stage Airflow (Electromechanical Condenser Only) B = Single Zone VAV (ReliaTel Condenser Only)

Table 4. Condenser and air handler pairing instructions (See document SS-SVN016A-EN)

Condenser (model # digit)	Air Handler		Instructions
	Type	Supply Fan Type (model # digit)	
Odyssey Electromechanical (Digit 15 = E)	Odyssey Symbio	Constant Volume (Digit 15 = 1)	Pairing F, D or G require wire harness kit WIR010190 (required) and WIR010185 (optional) to connect Air Handler Relay Board to VFD.
		2-Speed Airflow (Digit 15 = C)	
		Single Zone VAV (Digit 15 = D)	
Odyssey ReliaTel (Digit 15 = R)	Odyssey Symbio	Constant Volume (Digit 15 = 1)	Pairing F, D or G require wire harness kit WIR010190 (required) and WIR010185 (optional) to connect Air Handler Relay Board to VFD.
		2-Speed Airflow (Digit 15 = C)	
		Single Zone VAV (Digit 15 = D)	

Condenser and Air Handler Pairings

Table 4. Condenser and air handler pairing instructions (continued) (See document SS-SVN016A-EN)

Condenser (model # digit)	Air Handler		Instructions
	Type	Supply Fan Type (model # digit)	
Odyssey Symbio (Digit 15 = S)	Odyssey Symbio	Constant Volume (Digit 15 = 1)	Install a shielded, twisted pair cable if the Air Handler has Electric Heat and/or requires Single Zone VAV operation (Trane IMC communication)
		2-Speed Airflow (Digit 15 = C)	Pairing G, H, and 2 will not have heat in defrost. Pairing G, H, and 2; electric heat will not operate if zone sensor installed, only with a thermostat Install a shielded, twisted pair cable if the Air Handler has Electric Heat and/or requires Single Zone VAV operation (Trane IMC communication)
		Single Zone VAV (Digit 15 = D)	Install a shielded, twisted pair cable if the Air Handler has Electric Heat and/or requires Single Zone VAV operation (Trane IMC communication) Install a shielded, twisted pair cable for Symbio Condenser control of the Air Handler supply fan VFD (Modbus communication)
	Odyssey Electromechanical	Constant Volume (Digit 15 = 0)	Pairing G, H, and 2 will not have heat in defrost.
		2-Speed Airflow (Digit 15 = A)	Pairing G, H, and 2; electric heat will not operate if zone sensor installed, only with a thermostat.
	Odyssey ReliaTel	Variable Speed, Single Zone VAV (Digit 15 = B)	Pairing G, H, and 2 will not have heat in defrost. Pairing G, H, and 2; electric heat will not operate if zone sensor installed, only with a thermostat. Install a shielded, twisted pair cable for Symbio Condenser control of the Air Handler supply fan VFD (Modbus communication)
			Pairing G, H, and 2; electric heat will not operate if zone sensor installed, only with a thermostat. Pairing F, D or G require wire harness kit WIR010190 (required) and WIR010185 (optional) to connect Air Handler Relay Board to VFD. This pairing requires the replacement of the RTOM module with a Symbio Relay Board (MOD03105) and that the VFD wires 81B, 82B, 93B, 94B and 94D be replaced with wire harness kit WIR010190 (required) and WIR010185 (optional). The Air Handler will operate as a 2-speed fan.
	Generic Air Handler	Constant Volume	
Two Symbio Condensers (2 condensers to 1 air handler)	Odyssey Electromechanical		